ParaDEVS Library

Start Here: a reading guide to every document

ParaDEVS Library workspace, Binghamton

Last updated October 11, 2026

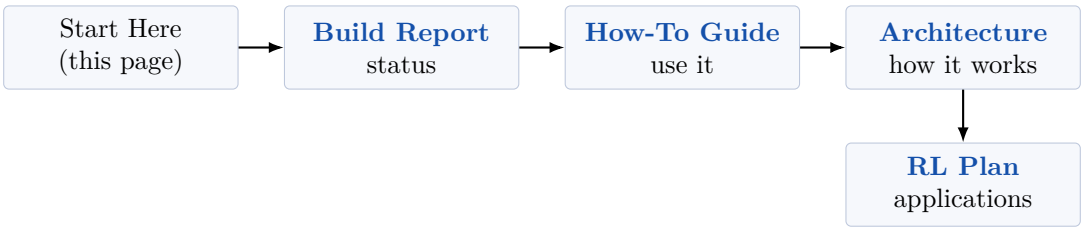
The project in three sentences. ParaDEVS is a Java library that runs a stochastic MS4 DEVS model as a *paratemporal* computation: whenever the model makes a random choice, the engine follows the possible outcomes and returns the resulting distribution. A model only has to get its random values from `Para.choose(label, law)`. Version 0 is built and verified (77 of 77 checks pass), and a reinforcement-learning track has started.

All titles below are clickable links that open the other PDFs. They point to the documents' current location; if the `ParaDEVS_Library` folder is moved, run `bash scripts/build_docs.sh` to refresh them.

1. Pick your path

I want to...	Read
know what exists, whether it works, and what is next	**Build Report**, Section 1 (one page)
write my first model and run it	**How-To Guide**: setup, then the scripted-model tutorial
bring a model over from MS4 Me	**How-To Guide**: the MS4 Me import tutorial
understand how the engine works inside	**Architecture Guide**
interpret a result (mass, residual, samples)	**Architecture Guide**: the result-interpretation section
understand "step" versus "draw"	**Architecture Guide**: the step-versus-draw section
see the RL plan and results	**RL Plan**
reproduce a number	**Build Report**, Appendix A
look up a term	**Build Report**, Appendix B (glossary)

2. Suggested reading order



New to the project: read left to right. Already using the library: jump straight to the document you need.

3. The documents

Build Report — 13 pages

Read it if: you want the overall picture: what was built, how it was verified, what went wrong, what is next.

Inside: one-page summary; starting point and decisions; the library's parts; verification against the paper, finance and baseball baselines; problems fixed; RL results in plain terms; limits; next steps and open decisions; timeline; commands to reproduce everything; glossary; file map.

File: `docs/report/ParaDEVS_Build_Report.pdf`

How-To Guide and Tutorial — 17 pages

Read it if: you want to use the library.

Inside: installation and VS Code setup; tutorial building a small DEVS model purely in Java; tutorial importing an MS4 Me model (what to bring over and the two edits needed); API reference; troubleshooting.

File: `docs/howto/ParaDEVS_HowTo.pdf`

Architecture and Interpretation Guide — 13 pages

Read it if: you want to know how it works, or what a number in a result means.

Inside: design principles (including step versus draw); the three layers and how they drive MS4; the life of a branch; merging and why it is safe; how to read results under each selection rule; which theory guarantees hold; extension points.

File: `docs/architecture/ParaDEVS_Architecture.pdf`

RL Applications Plan and Progress — 11 pages

Read it if: you are interested in reinforcement learning with the paratemporal engine.

Inside: where paratemporal fits in RL; Monte Carlo sampling with merging and its effect on variance; toy problems; Python baselines; options for connecting Java/MS4 to Python (including a DEVS message broker); phased plan R0–R5; progress and results so far, including the Java engine's n-step results with k branches.

File: `docs/rl/ParaDEVS_RL_Plan.pdf`

Control Spec: action selection with the engine — 12 pages

Read it if: you want to know how the engine chooses actions for value iteration, policy iteration and online control.

Inside: decision epochs and `Para.decide`; graph-mode planning (forward expansion, backward pass with max or policy average); discounting by simulation time; guarantees and the bias of sampled maxima; the API; phase plan 4a–4f.

File: `docs/rl/ParaDEVS_Control_Spec.pdf`

Hybrid Selection and Cost-Growth Study — 17 pages

Read it if: you want to choose which random draws to enumerate and which to sample, or how deep a lookahead can go.

Inside: what hybrid selection is; FrozenLake results per configuration; where the variance comes from; how work grows with lookahead, with and without merging; when exact expansion beats sampling; a rule of thumb.

File: `docs/rl/ParaDEVS_Hybrid_Study.pdf`

Engine Fast-Copy Proposal — 15 pages

Read it if: you want to understand why the engine replays steps, where its time goes, and how to make it faster.

Inside: why a half-finished step cannot be reused; a profile of each step; measured speed-ups of each option; the verification safeguard; a phased recommendation with decisions for you.

File: `docs/engine/ParaDEVS_FastCopy_Proposal.pdf`

4. Other material

Where	What
`ParaDEVS_Library/README.md`	Quick-start commands and a full map of the folder
`Do s/documents/paratemporal/`	The theory (authoritative): `main_paratemporal.tex`, `Paratemporal_Adaptation.tex`, `pt_reward_enumeration.py`
`ParaDEVS_Library/notes/`	Working notes from the first days (Markdown). Superseded by the PDFs above; kept for the decision log and to-do list in `03_scope_and_todo.md`
`ParaDEVS_Library/runs/`	Saved runs of the old engines, used as verification baselines
`ParaDEVS_Library/rl/experiments/results/`	Raw RL results (JSON, CSV, plots)

5. Keeping the documents up to date

Each PDF has a VS Code task (Terminal → Run Task, "compile the … PDF"). This guide and the build report are rebuilt, with fresh links, by `bash scripts/build_docs.sh`. All documents are LaTeX; the source sits next to each PDF. When a document is added, add a card for it here.